SDN Mininet based Simulation Project – Orange Problem

Goal:

The goal of this assignment is to implement an SDN-based solution using **Mininet and OpenFlow controller (Ryu/POX)** that demonstrates:

- Controller–switch interaction
- Flow rule design (match–action)
- Network behavior observation

This project must be done individually, not in a group.

Every project must include:

- Use Mininet for topology and use Ryu or POX controller
- Implement explicit flow rules (OpenFlow)
- Controller Logic must handle packet_in events and match+action logic.
- Functional behavior must be clearly demonstrated (Using tools - Wireshark, iperf etc).

For Testing & Validation:

- Demonstrate **working behavior clearly**
- Show **at least 2 test scenarios**, e.g.:
  - Allowed vs blocked
  - Normal vs failure
- Include **basic validation (or regression if applicable)**

Final Deliverable

Students must submit:

Final demo code should be shared in GitHub repo with complete documentation.

1. **Working Demonstration**
   - Live demo in Mininet
   - Functional correctness
2. **Source Code on GitHub repo (final demo code), make it public**
   - Clean and modular
   - Proper comments
3. **README Documentation on GitHub repo**
   - Problem statement
   - Setup/Execution steps
   - Expected output
4. **Proof of Execution (can be added in ReadMe on student's GitHub repo)**
   - Screenshots / logs (using Wireshark or other applicable tools):
     - Flow tables

UE24CS252B: Computer Networks

Topic Allotted: ***QoS Priority Controller***

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Sem: 4

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> X X X h5 X
h2 -> X X X h5 X
h3 -> X X X h5 X
h4 -> X X X h5 X
h5 -> h1 h2 h3 h4 X
h6 -> X X X X X
*** Results: 73% dropped (8/30 received)
```

Case 1: Pingall

This case specifically tells us about which host to host communication is allowed and which is not allowed also can see that one of the hosts in this configuration is blocked from every other host.

Controller (RYU):

```
arin@ArinSingh:~/Downloads$ sudo docker run -it --rm -p 6653:6653 -v $(pwd):/app -w /app -e PYTHONPATH=/app osrg/ryu ryu-manager controller
loading app controller
loading app ryu.controller.ofp_handler
instantiating app controller of FinalController
[INIT] Fixed Priorities assigned: h2(965) > h4(914) > h1(863) > h3(751)
instantiating app ryu.controller.ofp_handler of OFPHandler
```

Mininet:

```
arin@ArinSingh:~$ sudo mn --topo single,6 --controller=remote,ip=127.0.0.1,port=6653
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1) (h6, s1)
*** Configuring hosts
h1 h2 h3 h4 h5 h6
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

```
instantiating app ryu.controller.ofp_handler of OFPHandler
[METRIC] Latency (RTT): 24.78 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.1 -> 10.0.0.2)
[METRIC] Latency (RTT): 2.32 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.1 -> 10.0.0.3)
[METRIC] Latency (RTT): 2.87 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.1 -> 10.0.0.4)
[METRIC] Latency (RTT): 3.19 ms
[NORMAL] ALLOWED: 10.0.0.1 -> 10.0.0.5 | Priority: 10 | Total: 1
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[FAILURE] BLOCKED: Attacker involvement (10.0.0.1 -> 10.0.0.6)
[METRIC] Latency (RTT): 2.81 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.1 -> 10.0.0.2)
[METRIC] Latency (RTT): 1.83 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
[METRIC] Latency (RTT): 2.82 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.4)
[METRIC] Latency (RTT): 3.84 ms
[NORMAL] ALLOWED: 10.0.0.2 -> 10.0.0.5 | Priority: 10 | Total: 2
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[FAILURE] BLOCKED: Attacker involvement (10.0.0.2 -> 10.0.0.6)
[METRIC] Latency (RTT): 3.53 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.3 -> 10.0.0.1)
[METRIC] Latency (RTT): 4.07 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
[METRIC] Latency (RTT): 5.52 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.3 -> 10.0.0.4)
[METRIC] Latency (RTT): 2.62 ms
[NORMAL] ALLOWED: 10.0.0.3 -> 10.0.0.5 | Priority: 10 | Total: 3
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[FAILURE] BLOCKED: Attacker involvement (10.0.0.3 -> 10.0.0.6)
[METRIC] Latency (RTT): 4.76 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.4 -> 10.0.0.1)
[METRIC] Latency (RTT): 3.28 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.4 -> 10.0.0.2)
[METRIC] Latency (RTT): 3.48 ms
[FAILURE] BLOCKED: Isolation violation (10.0.0.3 -> 10.0.0.4)
[METRIC] Latency (RTT): 2.04 ms
[NORMAL] ALLOWED: 10.0.0.4 -> 10.0.0.5 | Priority: 10 | Total: 4
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[FAILURE] BLOCKED: Attacker involvement (10.0.0.4 -> 10.0.0.6)
[METRIC] Latency (RTT): 5.90 ms
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.1 | Priority: 863 | Total: 5
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[NORMAL] ALLOWED: 10.0.0.1 -> 10.0.0.5 | Priority: 10 | Total: 6
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
```

```
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.1 | Priority: 863 | Total: 5
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[NORMAL] ALLOWED: 10.0.0.1 -> 10.0.0.5 | Priority: 10 | Total: 6
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.2 | Priority: 965 | Total: 7
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[NORMAL] ALLOWED: 10.0.0.2 -> 10.0.0.5 | Priority: 10 | Total: 8
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
[METRIC] Latency (RTT): 2.91 ms
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.1)
[METRIC] Latency (RTT): 1.96 ms
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.2)
[METRIC] Latency (RTT): 3.95 ms
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.3)
[METRIC] Latency (RTT): 3.02 ms
[FAILURE] BLOCKED: Attacker involvement (10.0.0.4 -> 10.0.0.6)
[METRIC] Latency (RTT): 4.90 ms
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
[METRIC] Latency (RTT): 3.61 ms
```

Case 2: Ping from host (allowed) to host (server/host)

```
mininet> h5 ping 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=13.6 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=1.01 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.077 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.102 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.079 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.081 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=0.106 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=0.084 ms
64 bytes from 10.0.0.1: icmp_seq=9 ttl=64 time=0.084 ms
64 bytes from 10.0.0.1: icmp_seq=10 ttl=64 time=0.080 ms
64 bytes from 10.0.0.1: icmp_seq=11 ttl=64 time=0.100 ms
64 bytes from 10.0.0.1: icmp_seq=12 ttl=64 time=0.071 ms
^C
--- 10.0.0.1 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11352ms
rtt min/avg/max/mdev = 0.071/1.291/13.621/3.726 ms
```

```
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.1 | Priority: 863 | Total: 9
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
[NORMAL] ALLOWED: 10.0.0.1 -> 10.0.0.5 | Priority: 10 | Total: 10
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 0, 'HTTPS': 0}
```

Case 3: Ping from host to a blocked host (firewall application)

```
mininet> h5 ping h6
PING 10.0.0.6 (10.0.0.6) 56(84) bytes of data.
64 bytes from 10.0.0.6: icmp_seq=7 ttl=64 time=0.964 ms
64 bytes from 10.0.0.6: icmp_seq=8 ttl=64 time=0.076 ms
64 bytes from 10.0.0.6: icmp_seq=9 ttl=64 time=0.097 ms
64 bytes from 10.0.0.6: icmp_seq=10 ttl=64 time=0.080 ms
64 bytes from 10.0.0.6: icmp_seq=11 ttl=64 time=0.084 ms
64 bytes from 10.0.0.6: icmp_seq=12 ttl=64 time=0.115 ms
64 bytes from 10.0.0.6: icmp_seq=13 ttl=64 time=0.079 ms
64 bytes from 10.0.0.6: icmp_seq=14 ttl=64 time=0.086 ms
64 bytes from 10.0.0.6: icmp_seq=15 ttl=64 time=0.086 ms
64 bytes from 10.0.0.6: icmp_seq=16 ttl=64 time=0.108 ms
^C
--- 10.0.0.6 ping statistics ---
16 packets transmitted, 10 received, 37.5% packet loss, time 15967ms
rtt min/avg/max/mdev = 0.076/0.177/0.964/0.262 ms
```

```
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
[METRIC] Latency (RTT): 4.00 ms
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.5 -> 10.0.0.6)
```

1	0.000000000	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=1/256, ttl=64 (reply in 2)
2	0.013497204	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=1/256, ttl=64 (request in 1)
3	1.001371431	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=2/512, ttl=64 (reply in 4)
4	1.002344854	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=2/512, ttl=64 (request in 3)
5	2.026472671	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=3/768, ttl=64 (reply in 6)
6	2.026517602	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=3/768, ttl=64 (request in 5)
7	3.027171403	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=4/1024, ttl=64 (reply in 8)
8	3.027231598	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=4/1024, ttl=64 (request in 7)
9	4.058193560	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=5/1280, ttl=64 (reply in 10)
10	4.058237019	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=5/1280, ttl=64 (request in 9)
11	5.077449188	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=6/1536, ttl=64 (reply in 12)
12	5.077492647	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=6/1536, ttl=64 (request in 11)

Case 4: Ping host (server) to host (server) – (not allowed)

```
mininet> h2 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=1.83 ms
64 bytes from 10.0.0.3: icmp_seq=8 ttl=64 time=0.178 ms
64 bytes from 10.0.0.3: icmp_seq=9 ttl=64 time=0.072 ms
64 bytes from 10.0.0.3: icmp_seq=10 ttl=64 time=0.071 ms
64 bytes from 10.0.0.3: icmp_seq=11 ttl=64 time=0.067 ms
64 bytes from 10.0.0.3: icmp_seq=12 ttl=64 time=0.101 ms
64 bytes from 10.0.0.3: icmp_seq=13 ttl=64 time=0.095 ms
64 bytes from 10.0.0.3: icmp_seq=14 ttl=64 time=0.067 ms
64 bytes from 10.0.0.3: icmp_seq=15 ttl=64 time=0.076 ms
64 bytes from 10.0.0.3: icmp_seq=16 ttl=64 time=0.074 ms
64 bytes from 10.0.0.3: icmp_seq=17 ttl=64 time=0.082 ms
^C
--- 10.0.0.3 ping statistics ---
17 packets transmitted, 11 received, 35.2941% packet loss, time 16435ms
rtt min/avg/max/mdev = 0.067/0.247/1.834/0.502 ms
```

```
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
[FAILURE] BLOCKED: Isolation violation (10.0.0.2 -> 10.0.0.3)
```

1	0.000000000	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=1/256, ttl=64 (reply in 2)
2	0.013497204	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=1/256, ttl=64 (request in 1)
3	1.001371431	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=2/512, ttl=64 (reply in 4)
4	1.002344854	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=2/512, ttl=64 (request in 3)
5	2.026472671	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=3/768, ttl=64 (reply in 6)
6	2.026517602	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=3/768, ttl=64 (request in 5)
7	3.027171403	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=4/1024, ttl=64 (reply in 8)
8	3.027231598	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=4/1024, ttl=64 (request in 7)
9	4.058193560	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=5/1280, ttl=64 (reply in 10)
10	4.058237019	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=5/1280, ttl=64 (request in 9)
11	5.077449188	10.0.0.5	10.0.0.1	ICMP	98	Echo (ping) request id=0x3ad6, seq=6/1536, ttl=64 (reply in 12)
12	5.077492647	10.0.0.1	10.0.0.5	ICMP	98	Echo (ping) reply id=0x3ad6, seq=6/1536, ttl=64 (request in 11)
57	143.545791226	10.0.0.6	10.0.0.5	ICMP	98	Echo (ping) reply id=0xc90b, seq=13/3328, ttl=64 (request in 56)
58	144.735676529	10.0.0.5	10.0.0.6	ICMP	98	Echo (ping) request id=0xc90b, seq=14/3584, ttl=64 (reply in 59)
59	144.735725737	10.0.0.6	10.0.0.5	ICMP	98	Echo (ping) reply id=0xc90b, seq=14/3584, ttl=64 (request in 58)
60	145.747386695	10.0.0.5	10.0.0.6	ICMP	98	Echo (ping) request id=0xc90b, seq=15/3840, ttl=64 (reply in 61)
61	145.747435793	10.0.0.6	10.0.0.5	ICMP	98	Echo (ping) reply id=0xc90b, seq=15/3840, ttl=64 (request in 60)
62	146.828219031	10.0.0.5	10.0.0.6	ICMP	98	Echo (ping) request id=0xc90b, seq=16/4096, ttl=64 (reply in 63)
63	146.828285701	10.0.0.6	10.0.0.5	ICMP	98	Echo (ping) reply id=0xc90b, seq=16/4096, ttl=64 (request in 62)
64	150.105416557	fe80::2c9c:daff:feb…	ff02::2	ICMPv6	70	Router Solicitation from 2e:9c:da:b0:18:a9
65	150.105497542	fe80::d862:7fff:fe4…	ff02::2	ICMPv6	70	Router Solicitation from da:62:7f:49:f2:44
66	182.881327007	fe80::c1c:f2ff:fe13…	ff02::2	ICMPv6	70	Router Solicitation from 0e:1c:f2:13:0f:b2
67	301.333639074	10.0.0.2	10.0.0.5	ICMP	98	Echo (ping) request id=0x5beb, seq=1/256, ttl=64 (reply in 68)

Case 5: Ping blocked host to server host (h6 to h4):

```
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=7 ttl=64 time=0.927 ms
64 bytes from 10.0.0.4: icmp_seq=8 ttl=64 time=0.086 ms
64 bytes from 10.0.0.4: icmp_seq=9 ttl=64 time=0.102 ms
64 bytes from 10.0.0.4: icmp_seq=10 ttl=64 time=0.075 ms
64 bytes from 10.0.0.4: icmp_seq=11 ttl=64 time=0.079 ms
64 bytes from 10.0.0.4: icmp_seq=12 ttl=64 time=0.096 ms
64 bytes from 10.0.0.4: icmp_seq=13 ttl=64 time=0.079 ms
64 bytes from 10.0.0.4: icmp_seq=14 ttl=64 time=0.076 ms
64 bytes from 10.0.0.4: icmp_seq=15 ttl=64 time=0.077 ms
64 bytes from 10.0.0.4: icmp_seq=16 ttl=64 time=0.088 ms
64 bytes from 10.0.0.4: icmp_seq=17 ttl=64 time=0.063 ms
64 bytes from 10.0.0.4: icmp_seq=18 ttl=64 time=0.102 ms
^C
--- 10.0.0.4 ping statistics ---
18 packets transmitted, 12 received, 33.3333% packet loss, time 17561ms
rtt min/avg/max/mdev = 0.063/0.154/0.927/0.233 ms
```

```
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.4)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.4)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.4)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.4)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.4)
[FAILURE] BLOCKED: Attacker involvement (10.0.0.6 -> 10.0.0.4)
```

1	0.000000000	10.0.0.6	10.0.0.4	ICMP	98	Echo (ping) request id=0x16b4, seq=1/256, ttl=64 (no response found!)
2	1.061818364	10.0.0.6	10.0.0.4	ICMP	98	Echo (ping) request id=0x16b4, seq=2/512, ttl=64 (no response found!)
3	2.091575945	10.0.0.6	10.0.0.4	ICMP	98	Echo (ping) request id=0x16b4, seq=3/768, ttl=64 (no response found!)
4	3.109945988	10.0.0.6	10.0.0.4	ICMP	98	Echo (ping) request id=0x16b4, seq=4/1024, ttl=64 (no response found!)
5	4.134026926	10.0.0.6	10.0.0.4	ICMP	98	Echo (ping) request id=0x16b4, seq=5/1280, ttl=64 (no response found!)

Case 6: Simultaneous Call to check priority handling of the packets: h2 --> h4 --> h1 --> h3

```
mininet> h1 python3 -m http.server 80 &
mininet> h1 python3 -m http.server 80 &
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
mininet> h2 python3 -m http.server 443 &
mininet> h2 python3 -m http.server 443 &
Serving HTTP on 0.0.0.0 port 443 (http://0.0.0.0:443/) ...
mininet> h3 python3 -m http.server 21 &
mininet> h3 python3 -m http.server 21 &
Serving HTTP on 0.0.0.0 port 21 (http://0.0.0.0:21/) ...
mininet> h4 python3 -m http.server 22 &
mininet> h4 python3 -m http.server 22 &
Serving HTTP on 0.0.0.0 port 22 (http://0.0.0.0:22/) ...
mininet> h5 bash -c "curl http://10.0.0.1:80 & curl http://10.0.0.2:443 & curl http://10.0.0.3:21 & curl http://10.0.0.4:22 &"
mininet>
```

UE24CS252B: Computer Networks

```
arin@ArinSingh:~$ sudo mn --topo single,6 --controller=remote,ip=127.0.0.1,port=6653
[sudo] password for arin:
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1) (h6, s1)
*** Configuring hosts
h1 h2 h3 h4 h5 h6
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h1 python3 -m http.server 80 &
mininet> h1 python3 -m http.server 80 &
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
mininet> h2 python3 -m http.server 443 &
mininet> h2 python3 -m http.server 443 &
Serving HTTP on 0.0.0.0 port 443 (http://0.0.0.0:443/) ...
mininet> h3 python3 -m http.server 21 &
mininet> h3 python3 -m http.server 21 &
Serving HTTP on 0.0.0.0 port 21 (http://0.0.0.0:21/) ...
mininet> h4 python3 -m http.server 22 &
mininet> h4 python3 -m http.server 22 &
Serving HTTP on 0.0.0.0 port 22 (http://0.0.0.0:22/) ...
mininet>
```

```
arin@ArinSingh:~$ sudo ovs-ofctl -O OpenFlow13 dump-flows s1 | sort -r -k 8
[sudo] password for arin:
 cookie=0x0, duration=6.809s, table=0, n_packets=6, n_bytes=448, idle_timeout=60, hard_timeout=120, priority=965,in_port=5,dl_src=06:e3:56:a1:aa:c4,dl_dst=c6:17:50:d0:ea:78 actions=o
utput:2
 cookie=0x0, duration=6.802s, table=0, n_packets=6, n_bytes=447, idle_timeout=60, hard_timeout=120, priority=914,in_port=5,dl_src=06:e3:56:a1:aa:c4,dl_dst=96:55:22:e1:41:92 actions=o
utput:4
 cookie=0x0, duration=6.805s, table=0, n_packets=7, n_bytes=510, idle_timeout=60, hard_timeout=120, priority=863,in_port=5,dl_src=06:e3:56:a1:aa:c4,dl_dst=56:bf:91:d0:39:c4 actions=o
utput:1
 cookie=0x0, duration=6.805s, table=0, n_packets=7, n_bytes=513, idle_timeout=60, hard_timeout=120, priority=751,in_port=5,dl_src=06:e3:56:a1:aa:c4,dl_dst=36:6d:b4:b1:80:88 actions=o
utput:3
 cookie=0x0, duration=6.833s, table=0, n_packets=8, n_bytes=1941, idle_timeout=60, hard_timeout=120, priority=10,in_port=4,dl_src=96:55:22:e1:41:92,dl_dst=06:e3:56:a1:aa:c4 actions=o
utput:5
 cookie=0x0, duration=6.830s, table=0, n_packets=6, n_bytes=1809, idle_timeout=60, hard_timeout=120, priority=10,in_port=3,dl_src=36:6d:b4:b1:80:88,dl_dst=06:e3:56:a1:aa:c4 actions=o
utput:5
 cookie=0x0, duration=6.830s, table=0, n_packets=7, n_bytes=1875, idle_timeout=60, hard_timeout=120, priority=10,in_port=2,dl_src=c6:17:50:d0:ea:78,dl_dst=06:e3:56:a1:aa:c4 actions=o
utput:5
 cookie=0x0, duration=6.830s, table=0, n_packets=6, n_bytes=1809, idle_timeout=60, hard_timeout=120, priority=10,in_port=1,dl_src=56:bf:91:d0:39:c4,dl_dst=06:e3:56:a1:aa:c4 actions=o
utput:5
OFPST_FLOW reply (OF1.3) (xid=0x2):
 cookie=0x0, duration=67.430s, table=0, n_packets=59, n_bytes=4266, priority=0 actions=CONTROLLER:65509
arin@ArinSingh:~$
```

```
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.4 | Priority: 914 | Total: 13
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 1, 'HTTPS': 0}
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.2 | Priority: 965 | Total: 14
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 0, 'SSH': 1, 'HTTPS': 1}
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.1 | Priority: 863 | Total: 15
[STATS] Current Statistics: {'FTP': 0, 'OTHER': 0, 'HTTP': 1, 'SSH': 1, 'HTTPS': 1}
[NORMAL] ALLOWED: 10.0.0.5 -> 10.0.0.3 | Priority: 751 | Total: 16
[STATS] Current Statistics: {'FTP': 1, 'OTHER': 0, 'HTTP': 1, 'SSH': 1, 'HTTPS': 1}
```

Case 7: **Flow Installation**

This tests:

- First packet → controller handles
- Next packets → switch handles (FAST)

```
arin@ArinSingh:~$ sudo mn --topo single,6 --controller=remote,ip=1
[sudo] password for arin:
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1) (h6, s1)
*** Configuring hosts
h1 h2 h3 h4 h5 h6
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h5 ping -c 1 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=16.3 ms

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 16.314/16.314/16.314/0.000 ms
mininet> h5 ping -c 1 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.36 ms

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2.363/2.363/2.363/0.000 ms
mininet>
```